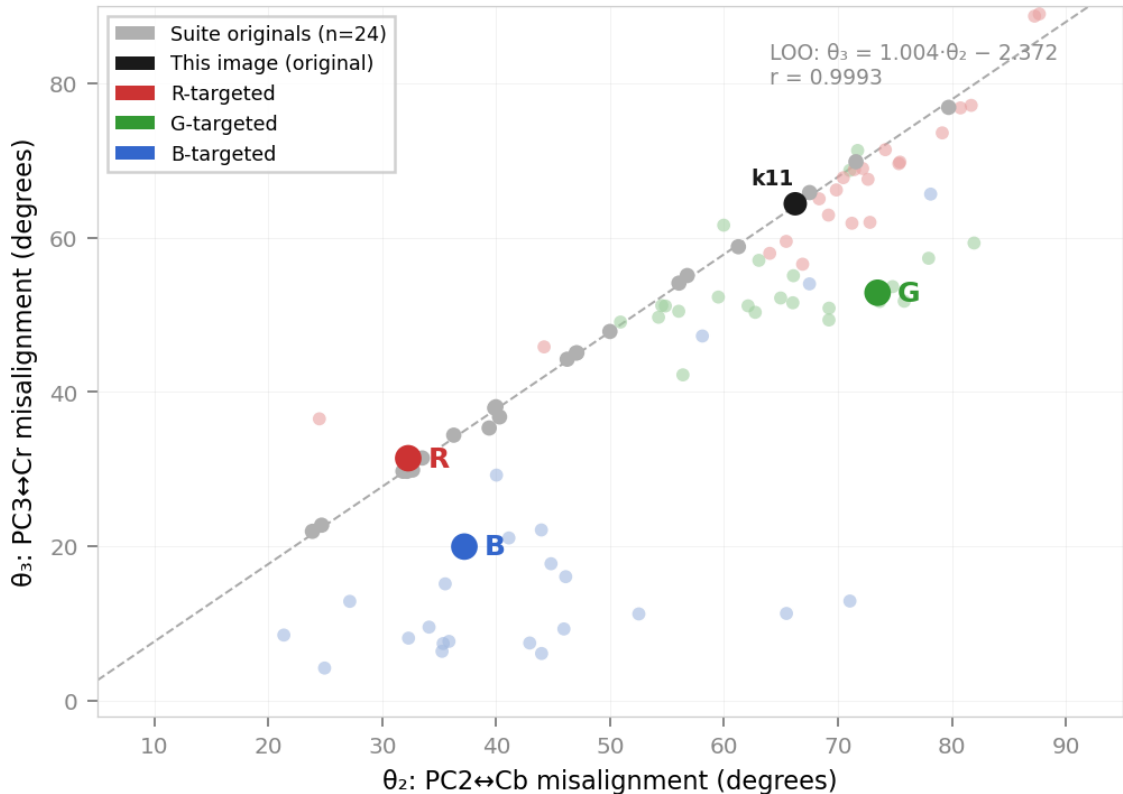


Compression Profile: KODIM11768x512 | 24-bit RGB | PNG (lossless)

sailboat at pier | Weakly One-Dimensional (PC1 = 91.56%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8225	0.7906	0.9744	0.863	66.25°	64.42°	+0.28°
R-targeted	0.0258	-0.0934	0.9386	0.353	32.27°	31.42°	+1.39°
G-targeted	-0.0361	0.9476	-0.0836	0.356	73.48°	52.88°	-18.52°
B-targeted	0.9216	-0.0019	0.0020	0.308	37.2°	19.96°	-15.02°

R-targeted isolates red (R-G, R-B < 0.09) while preserving G-B coupling at 0.939.
G-targeted isolates green (R-G, G-B < 0.08) while preserving R-B at 0.948.
B-targeted preserves R-G at 0.922 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM11 768x512 | 24-bit RGB | PNG (lossless)

sailboat at pier | Weakly One-Dimensional (PC1 = 91.56%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

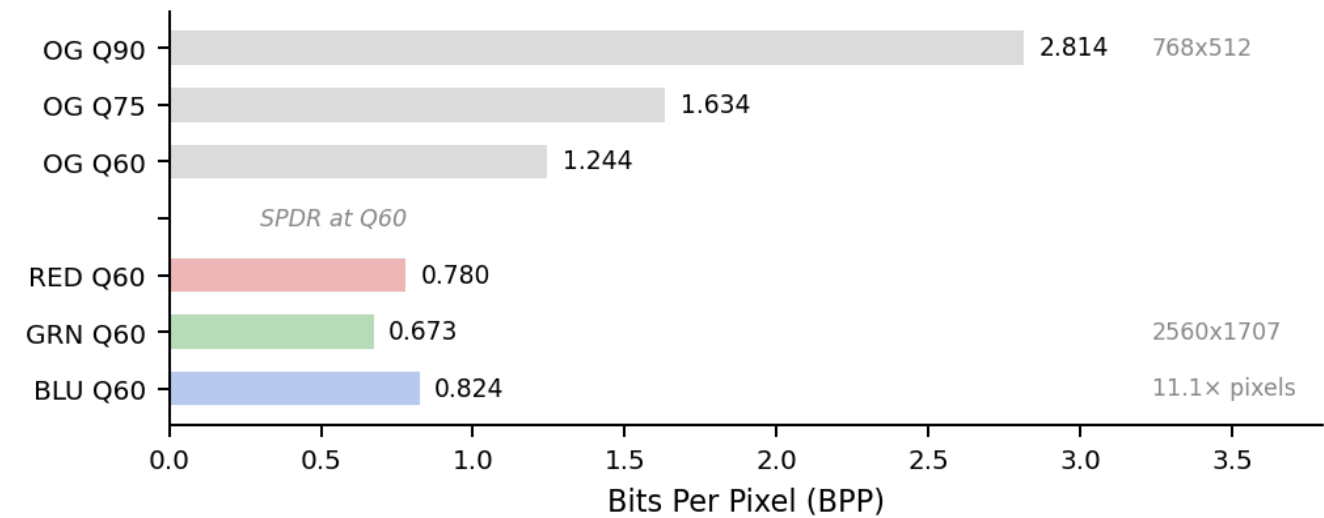
ORIGINAL (kodim11)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	606.5 KB	12.635	393,216
JPEG Q90	JPEG	768x512	135.1 KB	2.814	393,216
JPEG Q75	JPEG	768x512	78.4 KB	1.634	393,216
JPEG Q60	JPEG	768x512	59.7 KB	1.244	393,216

REDISTRIBUTED (kodim11 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	416.3 KB	0.780	4,369,920
GRN Q60	JPEG	2560x1707	359.1 KB	0.673	4,369,920
BLU Q60	JPEG	2560x1707	439.5 KB	0.824	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM11 768x512 | 24-bit RGB | PNG (lossless)

sailboat at pier | Weakly One-Dimensional (PC1 = 91.56%)

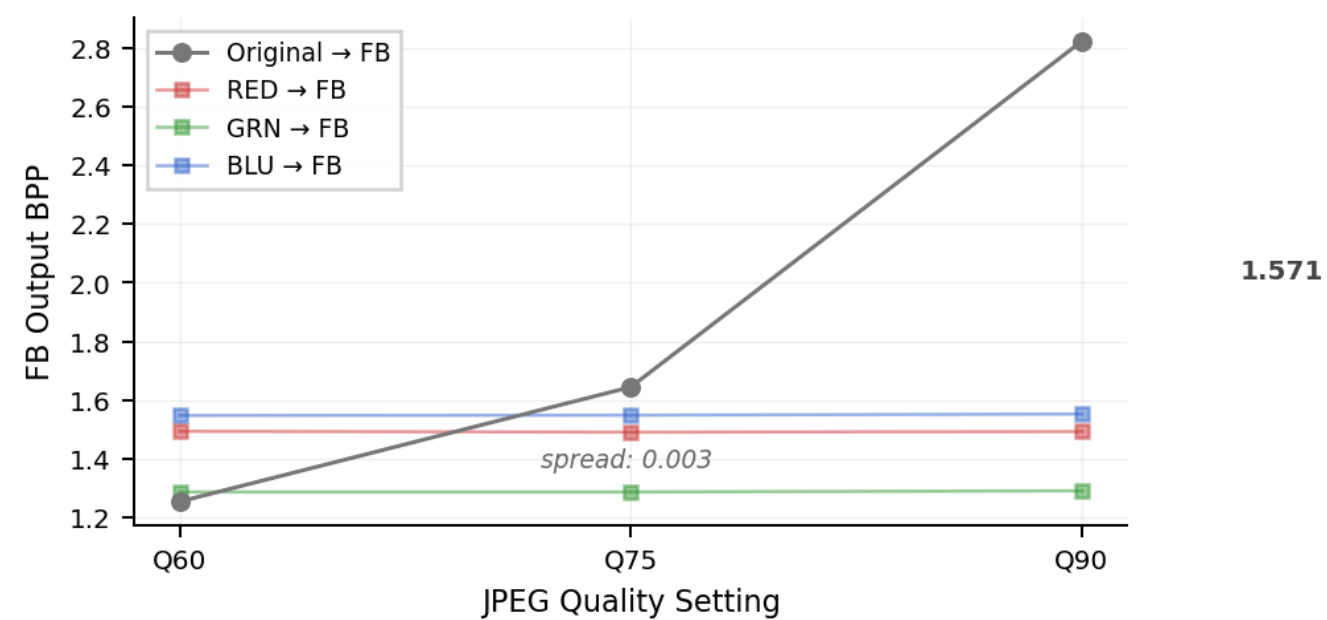
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.652	—
R-targeted → FB	2560x1707	2048x1366	1.492	43.7%
G-targeted → FB	2560x1707	2048x1366	1.290	51.4%
B-targeted → FB	2560x1707	2048x1366	1.552	41.5%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.402	47.1%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8610 vs 0.863 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.571 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.003 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim11

Best BPP (Q60, direct)	0.673	2560x1707, 4,369,920 px
Best BPP through FB	1.290	2048x1366 output
vs OG Q90 → FB	54.3%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216